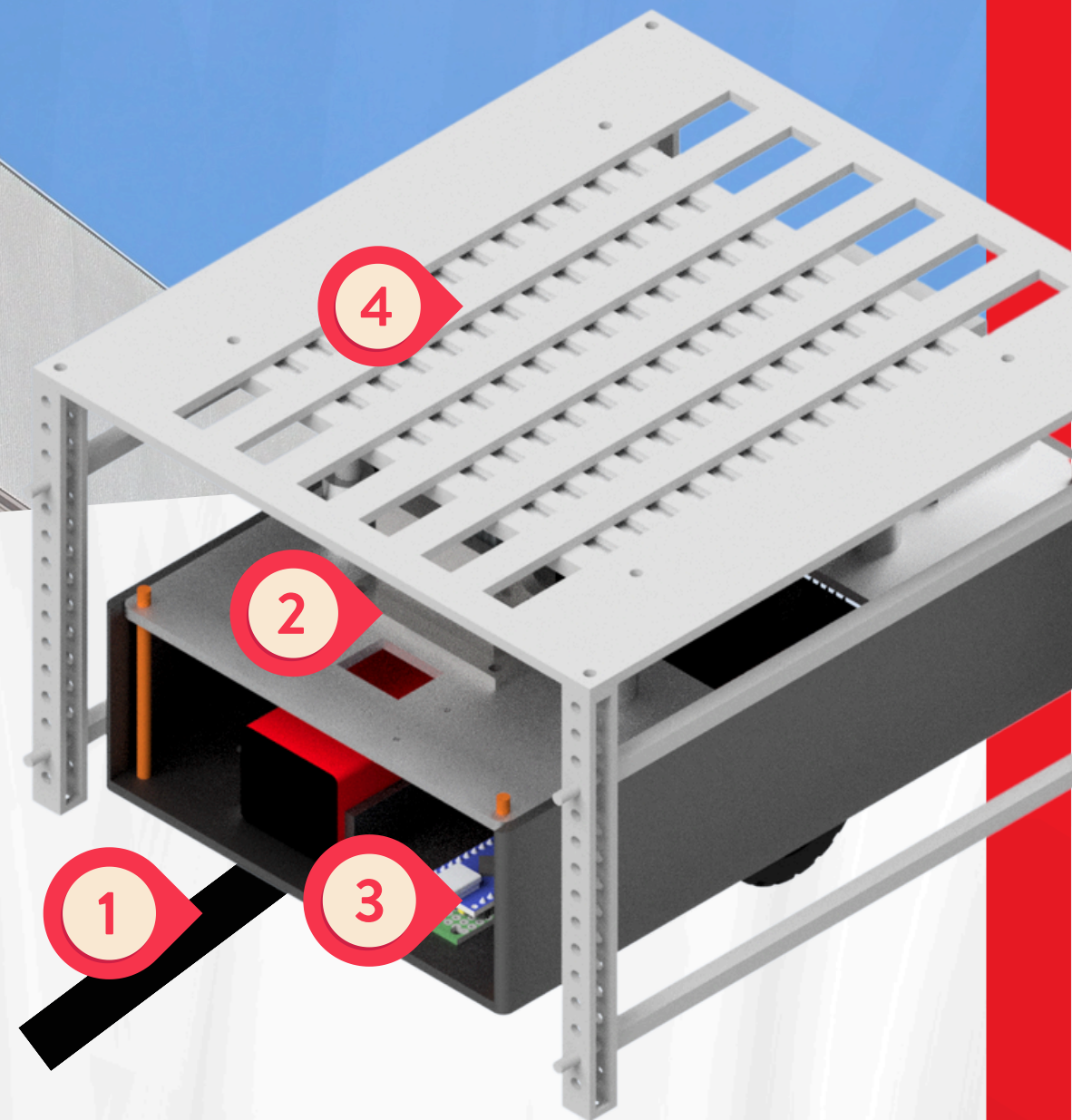


PATHOS

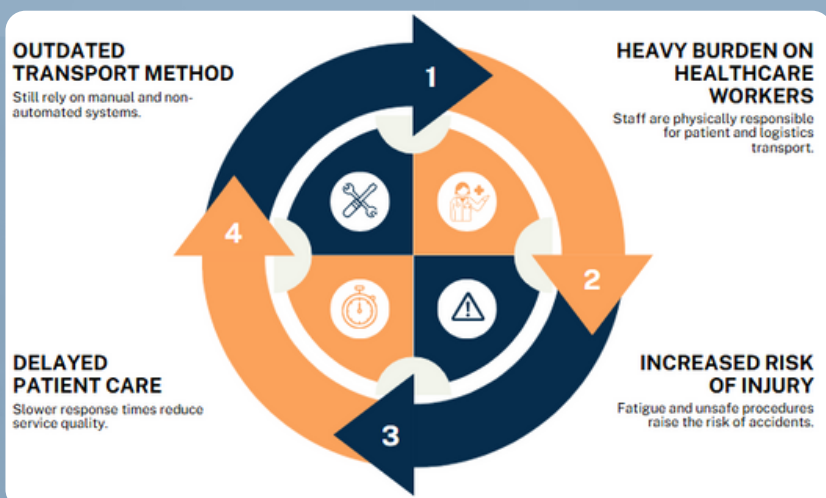
(Patient Transport & Hospital Operating System):

AGV-Based Smart Navigation for Efficient Patient and Supply Management

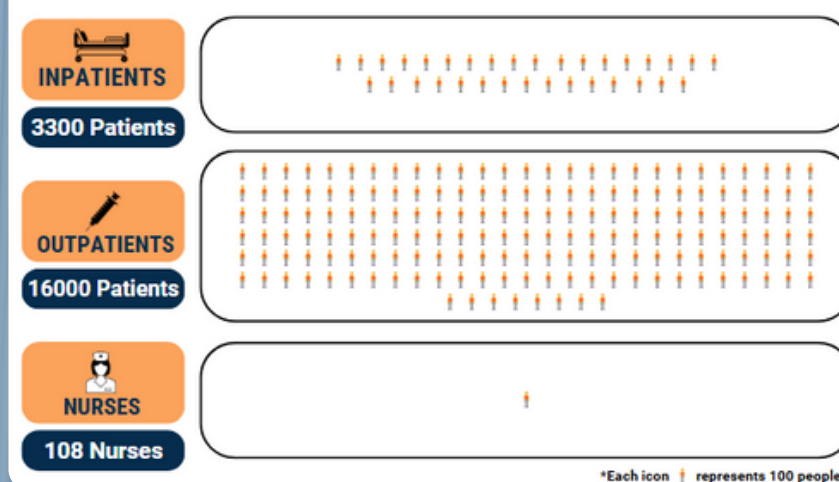


Background Problem

Hospitals continue to face challenges in patient transportation and medical logistics, with manual systems straining staff and increasing injury risks. RSUD Bangil, one of Indonesia's prominent public hospitals, serves thousands of patients monthly with only 108 nurses. This imbalance underscores the urgent need for an intelligent, automated solution to improve efficiency and support healthcare delivery.



Based on visitation data, Bangil Regional General Hospital (per-month) handles approximately:



Robot Feature

1. Smart Navigation System: Utilizes magnetic path sensors for guidance, with ultrasonic and infrared sensors for obstacle avoidance and RFID or QR code scanners for location tracking.
2. Robust Mechanical Structure: Features a strong frame with an automatic lifting mechanism and shock-absorbing wheels to ensure stable and comfortable patient transport.
3. Control and Integration System: Operates using microcontrollers like Raspberry Pi and ESP32 and integrates with the hospital management system for automated task scheduling.
4. Multifunctional Utility: Supports various functions including patient transport, delivery of medical supplies, and coordination among multiple AGVs, with a communication interface for staff interaction.

Impact



PATHOS enhances the quality of healthcare services by automating patient and medical logistics transport, reducing staff workload, minimizing injury risks, and enabling faster response times. This allows healthcare workers to focus more on critical care, ultimately improving patient outcomes and overall hospital efficiency.

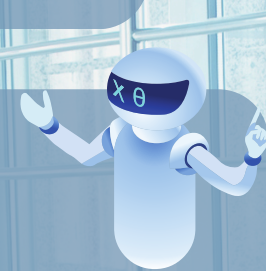


By integrating smart navigation, automation, and system connectivity, PATHOS brings innovation into hospital infrastructure. It promotes the development of sustainable, tech-driven solutions that modernize healthcare operations and set the foundation for future-ready, efficient hospital systems.

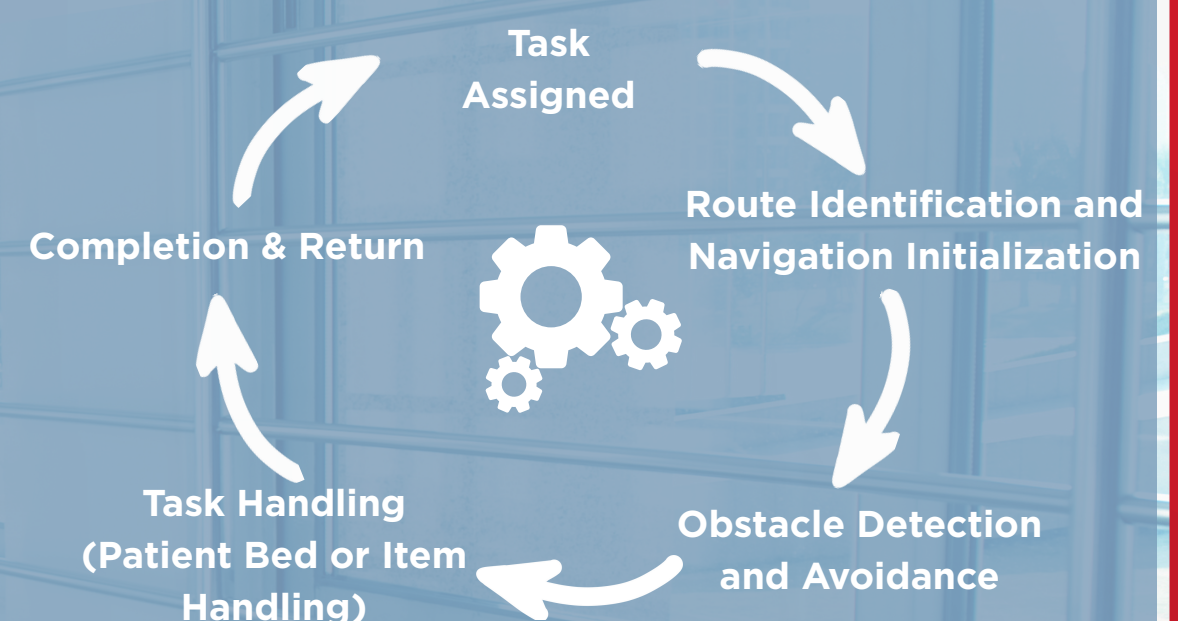


Conclusion

PATHOS offers an innovative approach to tackling the challenges of patient transportation and medical logistics in healthcare facilities. Through smart automation, seamless system integration, and multifunctional capabilities, it aims to reduce staff workload, enhance response times, and improve overall patient care. This solution holds great potential to support hospitals facing high patient volumes and limited human resources, paving the way for more efficient and future-ready healthcare services.



How it works



Demo Video:



<https://innopa.org/>
 @official_innopa

